CSE 1241 - COMPUTER PROGRAMMING I
Non-Graded Programming Assignment # 1

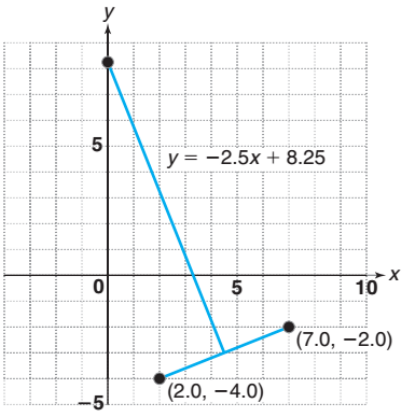
1. Suppose you want to develop a program that changes a given amount of money into smaller monetary units. The program lets the user enter an amount as a double value representing a total money in "TL" and "kuruş", and outputs a report listing the monetary equivalent in the maximum number of 200TLs, 100TLs, 50TLs, 20Tls, 10TLs, 5TLs, 1TLs, and 50Krs, 25Krs, 10Krs, 5Krs, 1Krs, in this order, to result in the minimum number of coins.

   An example output of your program should be like that:

```
Enter an amount in double, for example 11.56: 586.86
Your amount 586.86 consists of
    2 200TLs
    1 100TLs
    1 50TLs
    1 20TLs
    1 10TLs
    1 5TLs
    1 1TLs
    1 50Krs
    1 25Krs
    1 10Krs
    0 5Krs
    1 1Krs
```

   Hint: You can check the Case Study 2.17 "Counting Monetary Units" in your book!

2. Write a program that outputs the equation of the perpendicular bisector of the line segment between two points. Your program should
   a. prompt the appropriate messages and input the coordinates of the two points as (X_1, Y_1) and (X_2, Y_2) [for example, (2.0, -4.0) and (7.0, -2.0)];
   b. compute the slope of the line between those two points [$slope = \frac{Y_2 - Y_1}{X_2 - X_1}$];
   c. compute the coordinates of the midpoint of the line segment between the two points by averaging the two *x* coordinates and the two *y* coordinates; [$midpoint = \left(\frac{X_1 + X_2}{2}, \frac{Y_1 + Y_2}{2} \right)$]
   d. compute the slope of the perpendicular bisector by taking the negative reciprocal of the slope of the line segment; [$m = -slope^{-1}$];
   e. compute the y intercept of the perpendicular bisector (you now have the slope m of the bisector and the midpoint (x_{mid}, y_{mid}) on the bisector, so the y intercept is $y_{mid} - m * x_{mid}$);
   f. output with labels the original two points, and output in $\mathbf{y = mx + b}$ format the equation of the perpendicular bisector. [b is the y intercept of the perpendicular bisector found in previous step.]
      i. For example: y = -2.5x + 8.25
   g. The figure below illustrates the sample line segment mentioned above and its perpendicular bisector.



3. For this question, you will write a program that allows a car salesman to enter a customer's choices regarding a new car and from that information, itemize the costs of the choices and finally determine the total cost. The options and costs for the customer's choices are listed below.

   The program will prompt for several pieces of information. These data items will be used to determine the cost of the car. Some the responses may affect whether you ask questions later in the process. Once you've obtained all of the information, the total cost needs to be displayed.

Your program **will need** to accept responses with a ***mix*** of upper and lower case letters.

The Car Option Rules:

1. The dealership sells five car models with the following base prices:
   - a compact ($16,000)
   - a luxury ($39,000)
   - a midsize ($26,000)
   - a pickup ($19,500)
   - an SUV ($29,000)

2. There are six different features that might be added to a car model:
   - Protective undercoating
   - Automatic transmission
   - Sunroof
   - Multimedia package
   - Security package
   - Sports package

3. The following table presents the standard/optional/not available features and their prices for each car model.

Car model	Base price	Protective undercoating		Automatic transmission		Sunroof		Multimedia package		Security package		Sports package	
compact	$16,000	o	$125	-		-		-		o	$800	-	
luxury	$39,000	o	$125	s		s		o	$1250	s		-	
midsize	$26,000	o	$125	o	$2000	o	$1500	o	$750	o	$1000	o	5%
pickup	$19,500	o	$150	-		-		o	$750	-		-	
SUV	$29,000	o	$200	o	$2500	o	$2300	o	$1000	o	$1350	o	5%

- '***s***' indicates that this feature has already added to the car model by default (with no additional cost).
- '***o***' indicates that this feature might be added to the car model with an additional cost if the customer accepts.
- '**-**' indicates this feature is not available for this car model.

- *Protective undercoating* is an optional feature for each car model with different costs.
- *Automatic transmission* is a standard feature for a *luxury car*; however, it might be added as an optional feature for midsize car and SUV models with additional costs of $2000 and $2500, respectively. It is not available for compact car and pickup models.
- *Sunroof* is a standard feature for a *luxury car*; however, it might be added as an optional feature for midsize car and SUV models with additional costs of $1500 and $2300, respectively. It is not available for compact car and pickup models.
- *Multimedia package* is an optional feature for all car models except for compact cars with different costs.
- *Security package* is a standard feature for a *luxury car*; however, it might be added as an optional feature for compact, midsize and SUV models with additional costs of $800, $1000, and $1350 respectively. It is not available for pickup models.
- *Sports package* is an optional feature for midsize car and SUV models. The cost of the sports package is 5% of the base price of the model.

4. After retrieving the customer's choices, your program should add sales taxes to the total cost as the following:
    - Apply 35% OTV for the cars with total costs less than $35,000.
    - Apply 40% OTV for the cars with total costs higher than (or equal) $35,000.
    - Lastly, apply 18% KDV to the total cost after applying OTV.

5. Our car company also offers following advantageous packages for *luxury*, *midsize* and *SUV* models:
    - If a customer selects protective undercoating with multimedia package for a *luxury* car, then apply 5% discount to the total cost of these packages.
    - If a customer selects sunroof with multimedia package for a *midsize* car, then the total cost of these packages should be decreased to $2000.
    - If a customer selects security package with automatic transmission for a *midsize* car, then apply 10% discount to the total cost of these packages.
    - If a customer selects automatic transmission, sunroof, multimedia and security packages for an SUV model, then add protective undercoating for free.

    Here, example runs for different cases are provided.

Example run 1:

```
Welcome to Marmara Motor Cars
-----------------------------
Compact (C)
Luxury (L)
Midsize (M)
Pickup (P)
SUV (S)
What model of car do you want? (C, L, M, P, S): c
Do you want protective undercoating for $125.0 (Y/N)? Y
Do you want the security package for $800.0 (Y/N)? n

You ordered a Compact car with a base price of $16000.0
Your additional cost is $125.0
Sales Tax:  $9562.125
Total cost of car & options: $25687.125
Program ends...
```

Example run 2:

```
Welcome to Marmara Motor Cars
-----------------------------
Compact (C)
Luxury (L)
Midsize (M)
Pickup (P)
SUV (S)
What model of car do you want? (C, L, M, P, S): S
Do you want protective undercoating for $200.0 (Y/N)? n
Do you like to replace the normal transmission by an automatic
transmission for $2500.0 (Y/N)? y
Do you want sunroof for $2300.0 (Y/N)? Y
Do you want the multimedia package for $1000.0 (Y/N)? y
Do you want the security package for $1350.0 (Y/N)? Y
Do you want the sports package for an increased 5.0% This will increase
your cost by $1450.0 (Y/N)? N
You won protective undercoating for free!
```

```
You ordered an SUV with a base price of $29000
Your additional cost is $7150.0
Sales Tax:  $23569.8
Total cost of car & options: $59719.8
Program ends...
```

Example run 3:

```
Welcome to Marmara Motor Cars
-----------------------------
Compact (C)
Luxury (L)
Midsize (M)
Pickup (P)
SUV (S)
What model of car do you want? (C, L, M, P, S): L
Do you want protective undercoating for $125.0 (Y/N)? y
Do you want the multimedia package for $1250.0 (Y/N)? Y

You ordered a Luxury car with a base price of $39000.0
Your additional cost is $1306.25
Sales Tax:  $26279.675
Total cost of car & options: $66585.925
Program ends...
```